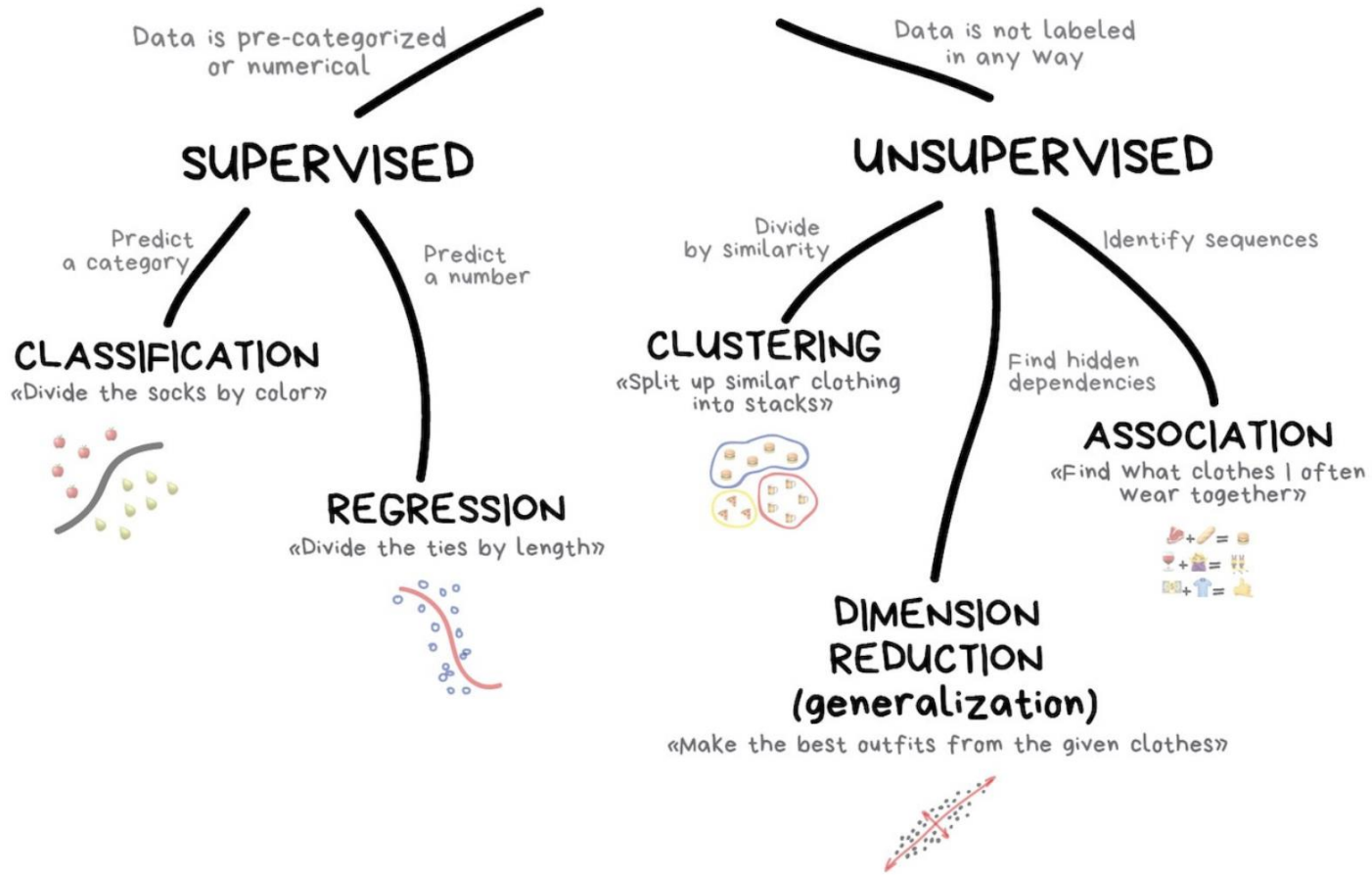
Predictive Analytics: Regression Analysis (การวิเคราะห์การถดถอย)

Assoc. Prof. Kitsana Waiyamai, Ph. D.

Based on Shmueli, Patel & Bruce

Dept. Of Computer Engineering
Faculty of Engineering, Kasetsart University
Bangkok, Thailand

Classical Machine Learning

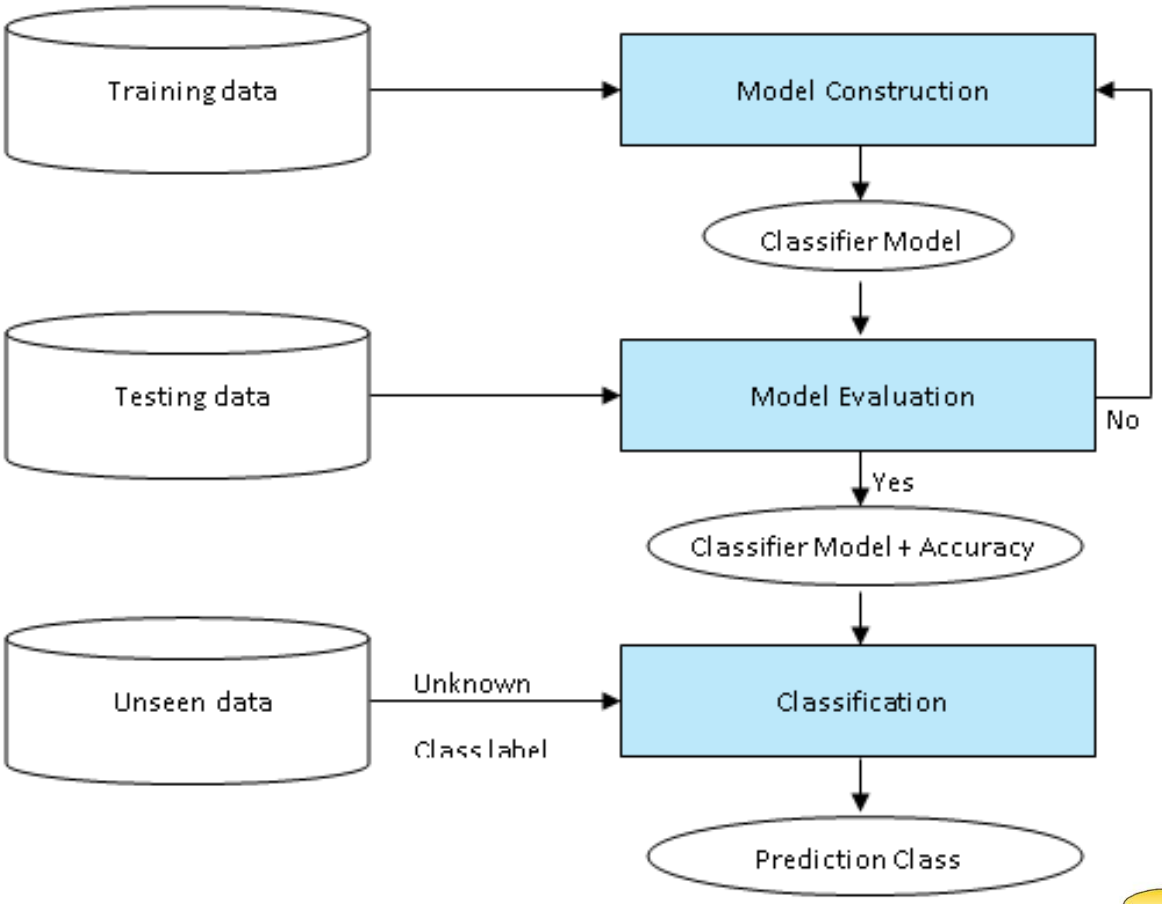


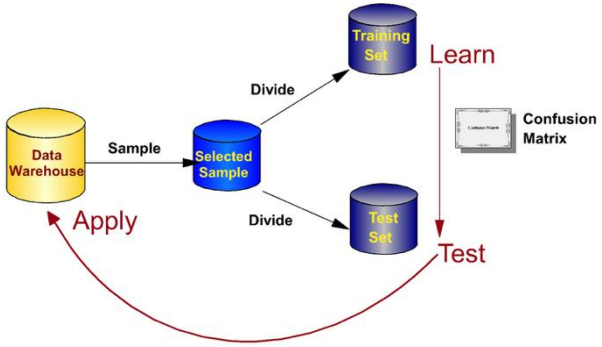
https://vas3k.com/blog/machine_learning/?ref=hn

Types of Learning Methods

- Supervised learning
  - We have INPUT and desired OUTPUT
  - Learns relation between INPUT and OUTPUT
  - We know the labels => Labeled Data
- Un-supervised learning
  - We have only INPUT but no OUTPUT
  - We do not really know what we are looking for
  - We want to discover hidden or un-known patterns
  - We do not know the classes => Un-Labeled Data
- Semi-Supervised learning
  - Use both INPUT and OUTPUT
  - Use both labeled and un-labeled data
- Reinforcement learning
  - We have GOAL and available ACTION
  - We want a model (agent) to GOAL achievement using available ACTION without human supervision
  - Kind of Trial and Error
  - Reward is associated with the action
  - Data is created during the learning step

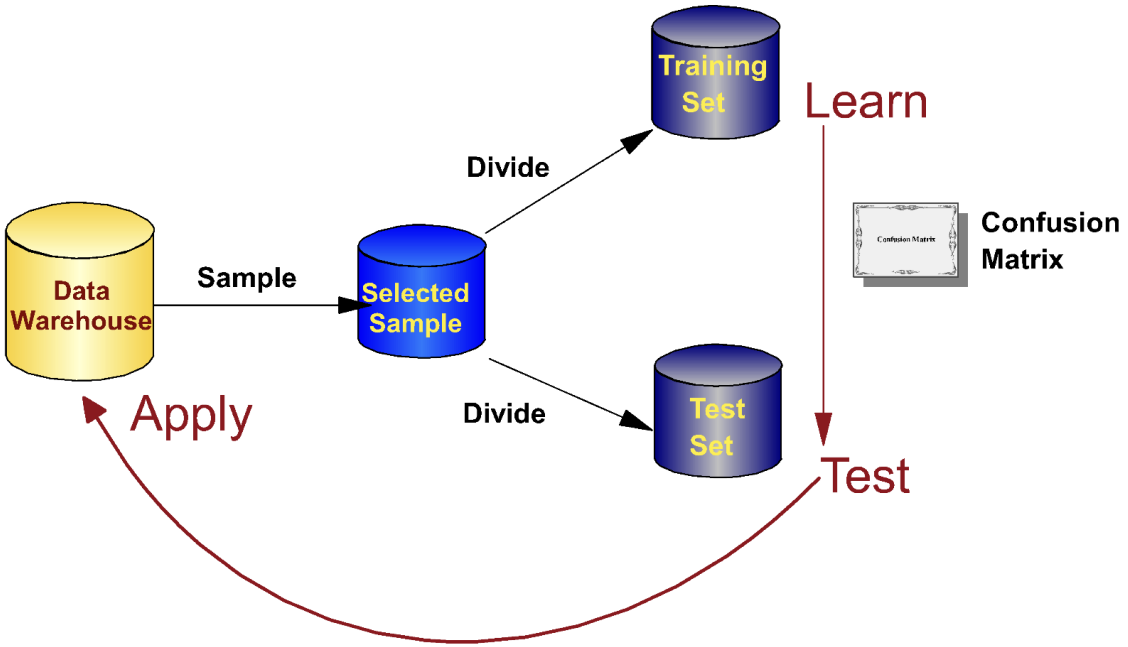
Steps of Supervised Learning





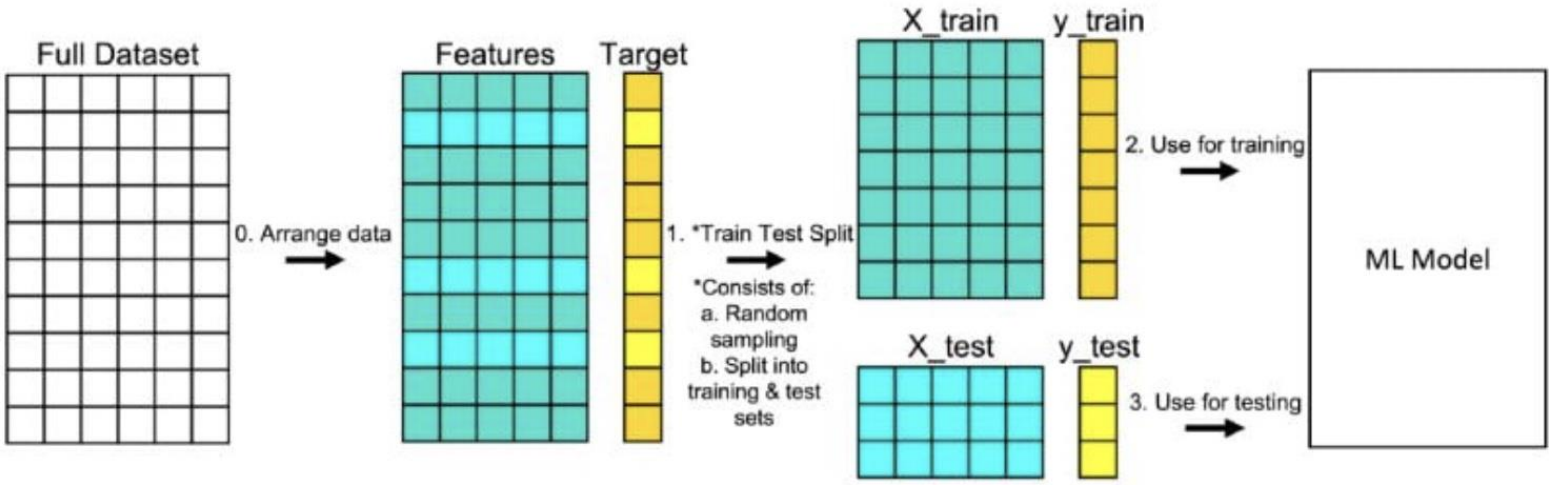
Hold-out divides the dataset into two parts

- Usually, the given data set is divided into training and test set, with training set used to build the model and test set used to validate it.
- A *test set* is used to determine the accuracy of the model.



Python train_test_split

Splitting the dataset into the Training set and Test set



Python train_test_split operator

Splitting the dataset into the Training set (80%) and Test set (20%)

```
[ ] from sklearn.model_selection import train_test_split
    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2, random_state = 1)
```

```
[ ] print(X_train)
```

```
[[0.0 0.0 1.0 38.77777777777778 52000.0]
 [0.0 1.0 0.0 40.0 63777.77777777778]
 [1.0 0.0 0.0 44.0 72000.0]
 [0.0 0.0 1.0 38.0 61000.0]
 [0.0 0.0 1.0 27.0 48000.0]
 [1.0 0.0 0.0 48.0 79000.0]
 [0.0 1.0 0.0 50.0 83000.0]
 [1.0 0.0 0.0 35.0 58000.0]]
```

```
[ ] print(X_test)
```

```
[[0.0 1.0 0.0 30.0 54000.0]
 [1.0 0.0 0.0 37.0 67000.0]]
```

```
[ ] print(y_train)
```

```
[0 1 0 0 1 1 0 1]
```

```
[ ] print(y_test)
```

```
[0 1]
```

```
[ ] print(X)
```

```
[[1.0 0.0 0.0 44.0 72000.0]
 [0.0 0.0 1.0 27.0 48000.0]
 [0.0 1.0 0.0 30.0 54000.0]
 [0.0 0.0 1.0 38.0 61000.0]
 [0.0 1.0 0.0 40.0 63777.77777777778]
 [1.0 0.0 0.0 35.0 58000.0]
 [0.0 0.0 1.0 38.77777777777778 52000.0]
 [1.0 0.0 0.0 48.0 79000.0]
 [0.0 1.0 0.0 50.0 83000.0]
 [1.0 0.0 0.0 37.0 67000.0]]
```

Regression Analysis (Prediction)

- Regression is a specific type of **supervised learning** that focuses on **continuous outcomes**.
- Predicting house prices
  - Regression models capture the underlying relationship between
    - house features (**independent variables**) and
    - price (**dependent variable**).
  - Enabling us to not only estimate the price of a new house but also understand **how different features influence the price**.

Regression Analysis

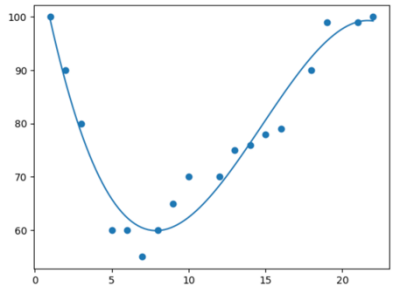
- **A statistical technique that analyzes the "relationship" between variables => Predict value of Y using value of X**
- It helps us:
  - Predict **numerical values**
  - Understand **relationships between variables**
  - Quantify the effect of each feature on the outcome
- **Simple Examples**
  - Predict salary from years of experience
  - Predict house price from area size (sq.m.)
  - Predict exam score from hours of study
  - Predict sales from marketing budget
- Mathematically, we try to estimate:

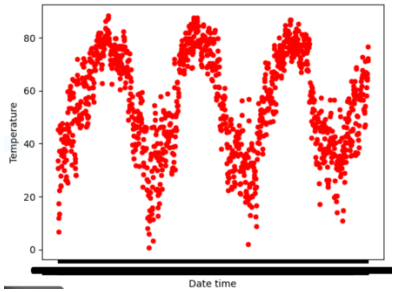
$$Y = f(X) + \varepsilon$$

- Y = **Dependent variable** (value being predicted, target, response, label)
- f(X) = function describing the relationship
- X = **Independent variable** (predictor, feature)
- ε = **error** term (noise)

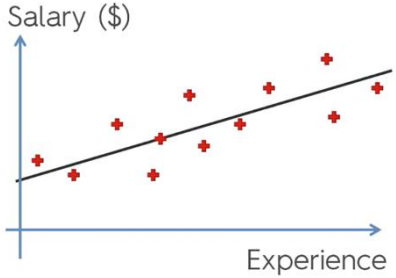
Linear Regression Analysis

- Regression models, both linear and non-linear, are used for predicting a numerical value
- Regression techniques:
  - **Simple Linear** Regression
  - **Multiple Linear** Regression
  - Polynomial Regression
  - Decision Tree Regression
  - Random Forest Regression
  - Neural Networks
  - Etc.





- **Linear Linear Regression**: a regression model where the relationship between input variables and output is **linear in parameters**.
- **Linear** => Y can be expressed in terms of **linear combination of the coefficients** of X
  - Straight-line relationship
  - Parameters appear linearly
  - Easy to interpret

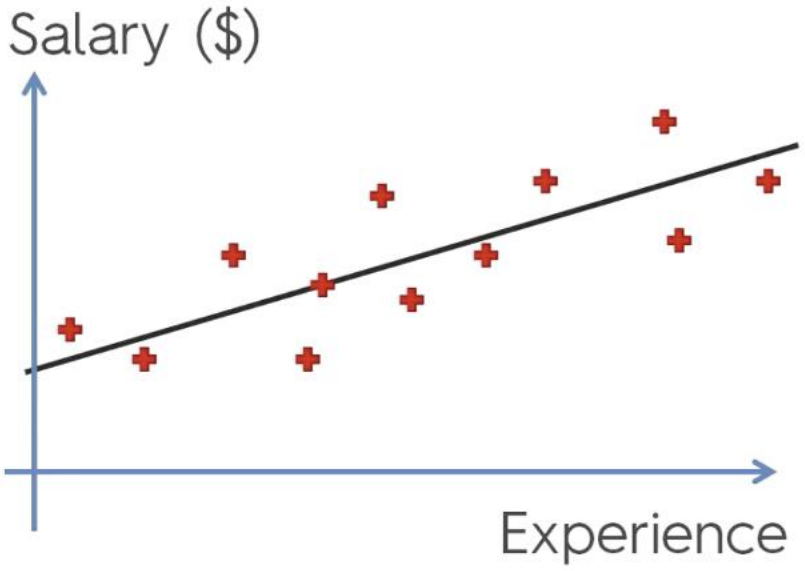


Linear Regression Analysis

- Simple Linear Regression
  - X => Y
  - Y Linearly depends on X

- Linear regression analysis estimate the relationship between an **independent variable (Experience)** and a **dependent variable (Salary)** by

  ⇒ fitting a straight line to the data.
  ⇒ Draw the best straight line to the data.

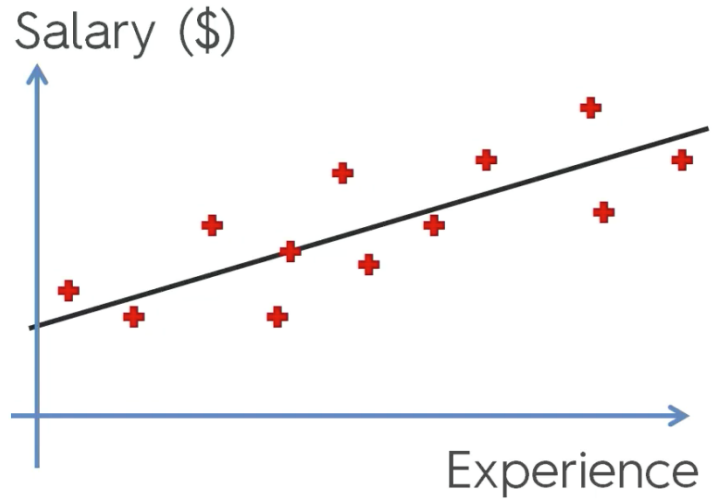


YearsExperience	Salary
1.1	39343
1.3	46205
1.5	37731
2	43525
2.2	39891
2.9	56642
3	60150
3.2	54445
3.2	64445
3.7	57189
3.9	63218
4	55794

Linear Regression Analysis

Y can be expressed in terms of linear combination of the coefficients of X

Simple Linear Regression:

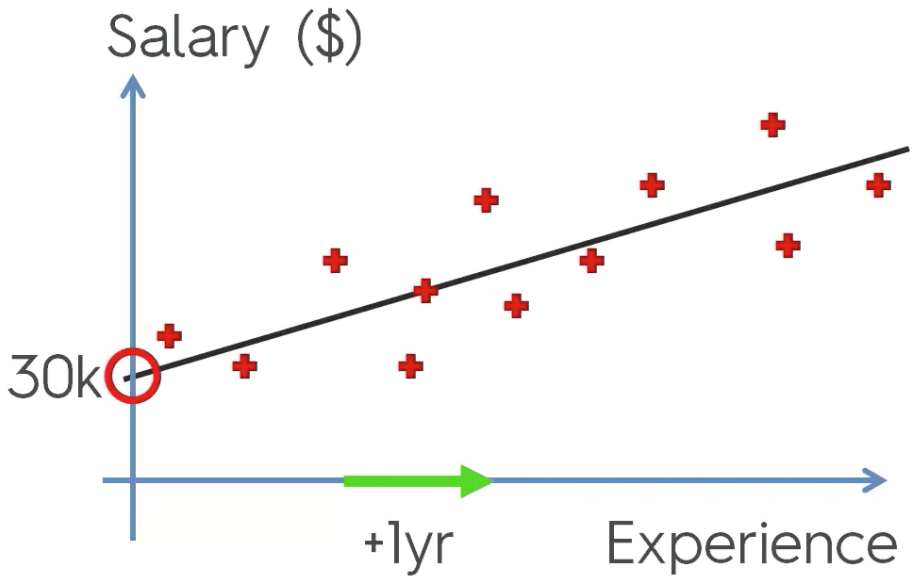


$y = b_0 + b_1 * x$

$\text{Salary} = b_0 + b_1 * \text{Experience}$

Simple Linear Regression

Simple Linear Regression:



$$y = b_0 + b_1 * x$$

Salary = b_0 + b_1 *Experience

Simple Linear Relationship

Linear relationship between **two variables** is stated as

$$y = \beta_0 + \beta_1 x$$

This is the general equation for a line (for the time being forget ε)

y : The dependent variable (what you want to predict)

x : The independent variable (what affects y)

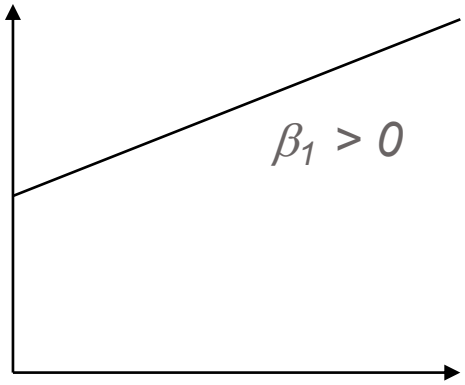
β_0 : The intersection with y axis, y-intercept (value of y when x = 0)

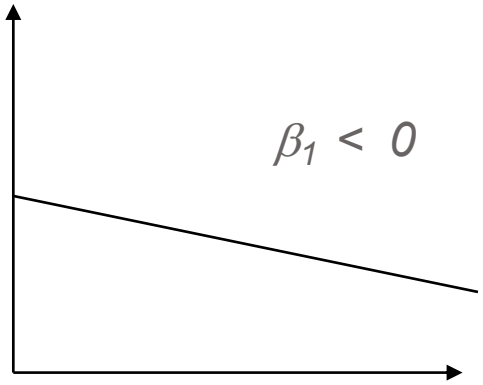
β_1 : The slope (indicates direction and strength of relationship)

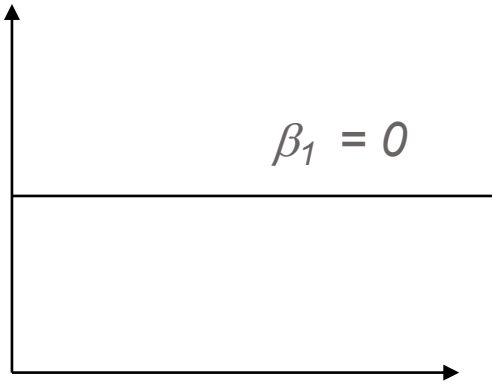
Simple Linear Regression Equation

$y = \beta_0 + \beta_1 x$

β_0 is the intersection with y axis, β_1 is the slope.







Multiple Linear Regression

Simple Linear Regression

$$y = b_0 + b_1 * x_1$$

Multiple Linear Regression

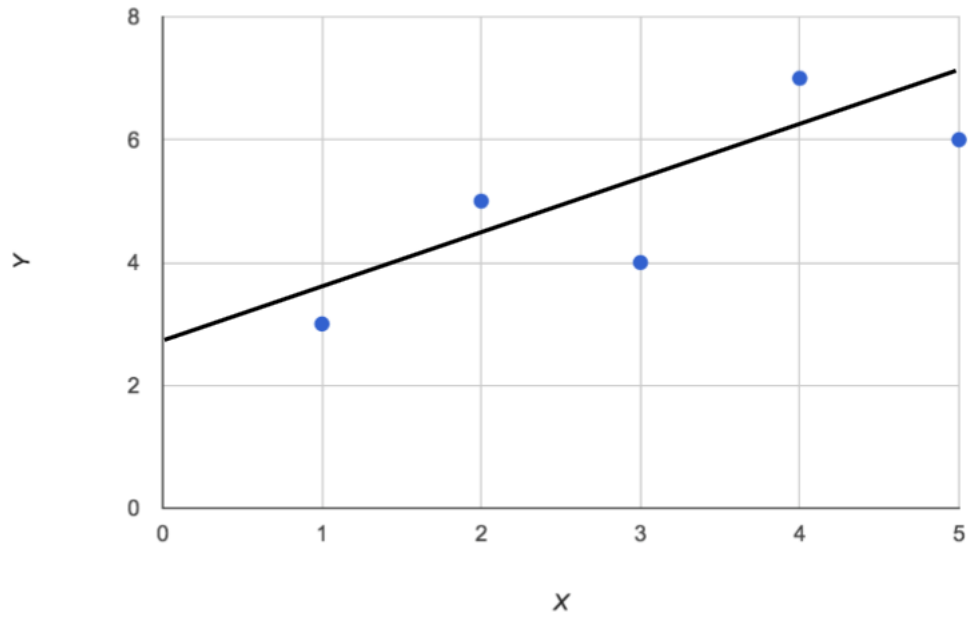
Dependent variable (DV) — Independent variables (IVs)

$$y = b_0 + b_1 * x_1 + b_2 * x_2 + \ldots + b_n * x_n$$

Simple Linear Regression Equation

Functions $f(x)=y$ that are lines: $y = a\,x + b$

$$y = 0.8\,x + 2.6$$



Use of linear regression for predictive modeling

1. Define the equation:
   Start with a Simple Linear Equation: $y = \beta_0 + \beta_1 x$

2. Linear Regression Model Learning:
   Based on your training data, estimate the values of β_1 and β_0 that make the line best fit the data points.

3. Evaluate Model on testing data:
   Assess the model's performance using measures like R-squared, p-value.

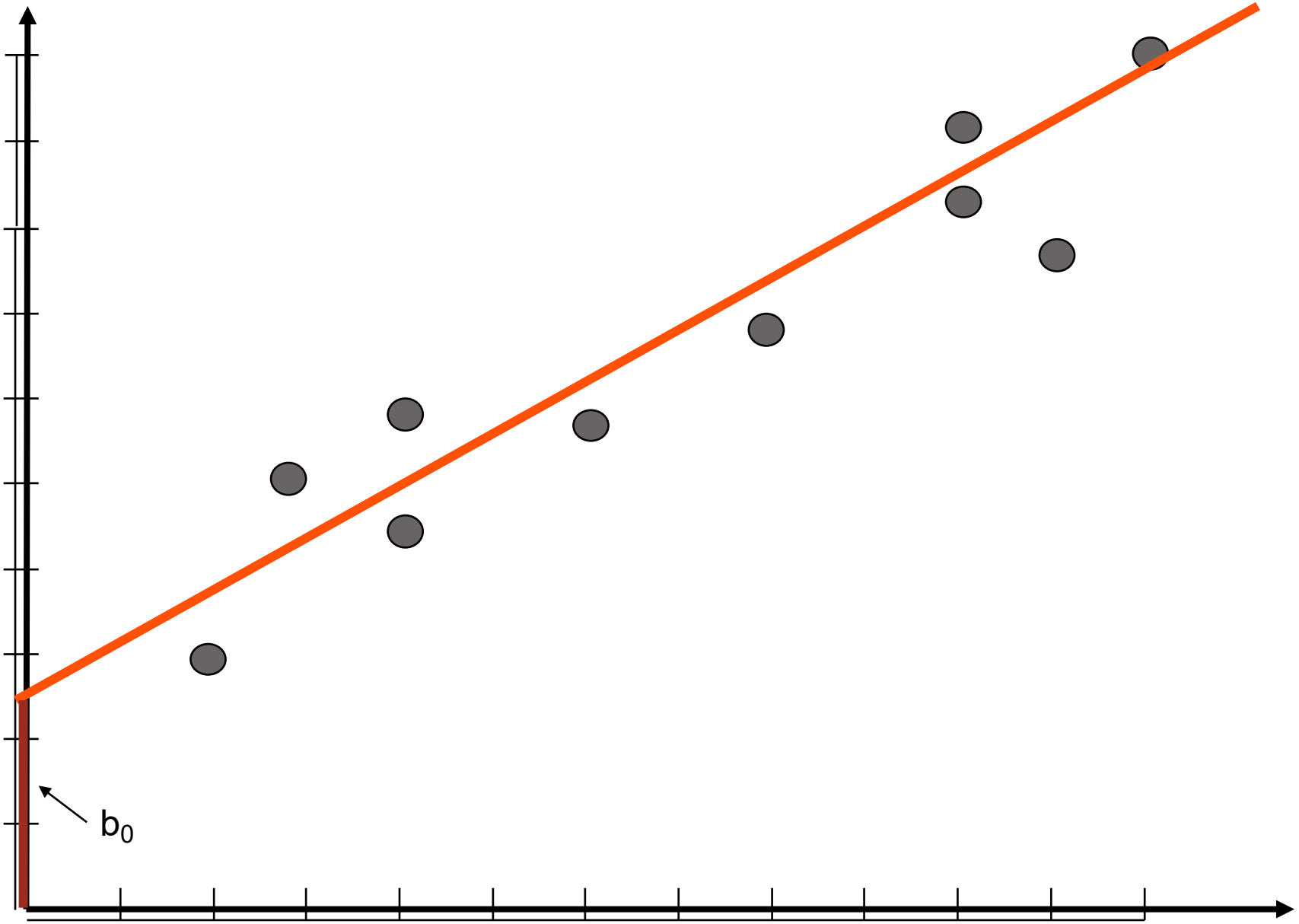
4. Make predictions:
   Use the fitted model to predict y values for new x values.

Predictive Modeling

Goal: predict target values

- Model Goal: Optimize predictive accuracy
- Train model on training data
- Assess performance on test (hold-out) data
- Explaining role of predictors is very useful

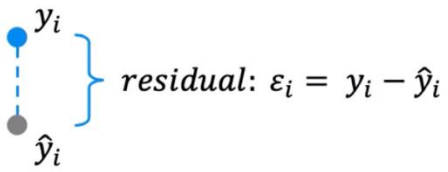
Graphical - Judgmental Solution



Learning to Build a Linear Regression Equation Model

The Ordinary Least Square (OLS) Method

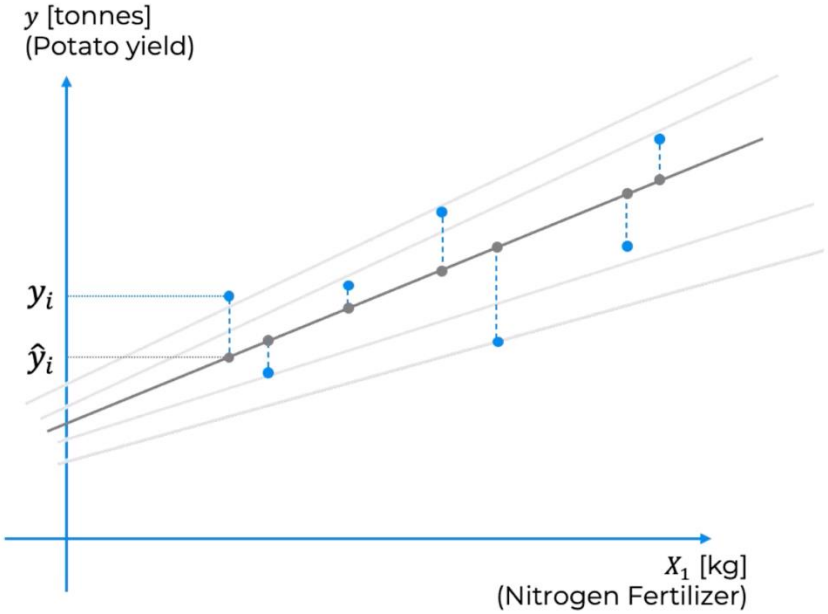
Ordinary Least Squares:



$\hat{y} = b_0 + b_1 X_1$

b_0, b_1 such that:

$SUM(y_i - \hat{y}_i)^2$ is minimized



The Least Square Method

Our judgmental approach and our eyes tried to minimize the difference between observed values and values obtained on the estimated regression line.

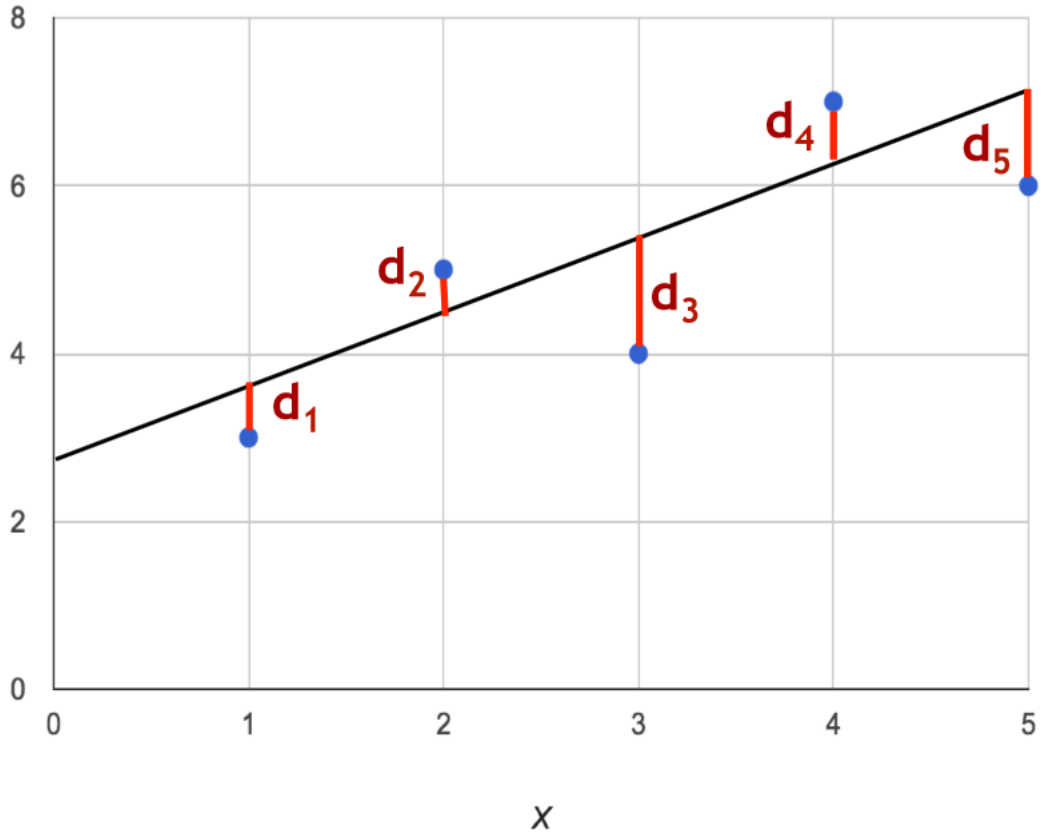
We may implement the same approach in algebra by

Minimize Sum of the square of the difference between observed values and values on the regression line.

The Least Square Method.

Calculate Linear Regression

Method of least squares



$$SSE = d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2$$

The Least Square Method

y_i	x_i	$\hat{y}_i$
y_1	x_1	$b_0 + b_1 x_1$
y_2	x_2	$b_0 + b_1 x_2$
y_3	x_3	$b_0 + b_1 x_3$
.	.	.
.	.	.
y_n	x_n	$b_0 + b_1 x_n$

$$\textbf{Min} \qquad Z = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$\textbf{Min} \qquad Z = \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i)^2$$

Classic Minimization

$$Min \qquad Z = \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i)^2$$

We want to minimize this function with respect to b_0 and b_1
This is a classic optimization problem.

We may remember from high school algebra that to find the minimum value we should get the derivative and set it equal to zero.

B_0 and b_1

$$b_1 = \frac{\sum xy - (\sum x \sum y)/n}{\sum x^2 - (\sum x)^2/n}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

Coefficient of Determination: r^2

Question : **How well** does the estimated **regression line fits the data.**

r^2 : **Coefficient of determination** is a measure for Goodness of Fit.
How well the data fit the regression model

Goodness of Fit of the estimated regression line to the data.

$1 \geq r^2 \geq 0$

$r^2 = .9027$

In other words, 90% of variations in y can be explained by the regression line.

R^2 tells you what percentage of the variation in the target can be explained by the predictors.

r^2 : Coefficient of Determination determines the proportion of variance in the dependent variable that can be explained by the independent variables.

Coefficient of Determination: r^2

Question : **How well** does the estimated **regression line fits the data.**

Coefficient of determination is a measure for Goodness of Fit.
Goodness of Fit of the estimated regression line to the data.

$(y_i - \hat{y}_i)$ is called **residual.**

It is the error in using **$\hat{y}_i$** to estimate y_i.

SSE = Sum of Squared Errors = sum of the squared residuals.
SST = Sum of Squared Total = sum of the squared deviations of the dependent variable from its mean

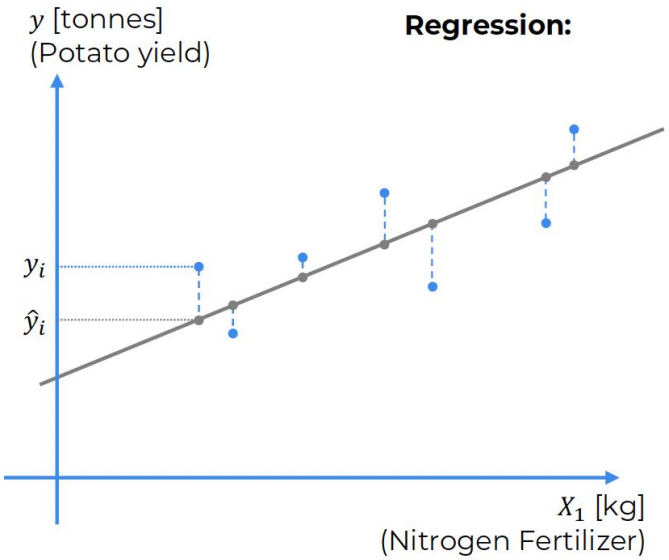
$R^2 = 1 - (SSE / SST)$

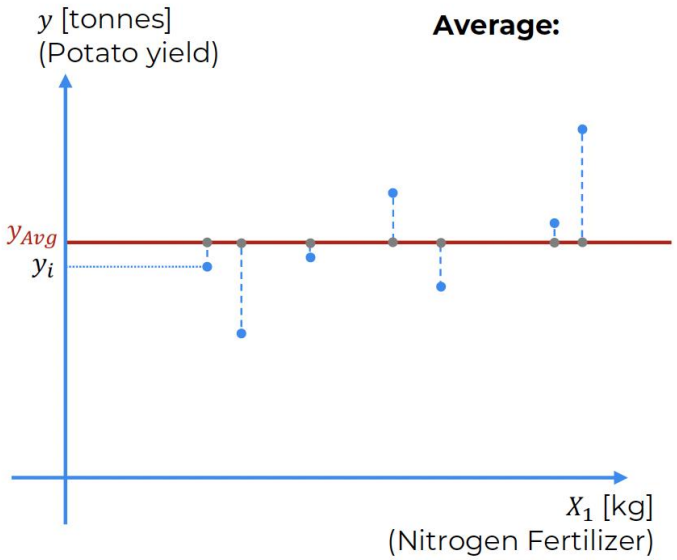
$$R^2 = 1 - \left(\Sigma (y_i - \hat{y}_i)^2 / \Sigma (y_i - \bar{y})^2\right.$$

Where:

y_i: Actual value
$\hat{y}_i$: Predicted value
$\bar{y}$: Mean of the actual values
n: Number of data points

Coefficient of Determination: r^2





$$SS_{res} = SUM(y_i - \hat{y}_i)^2$$

$$SS_{tot} = SUM(y_i - y_{avg})^2$$

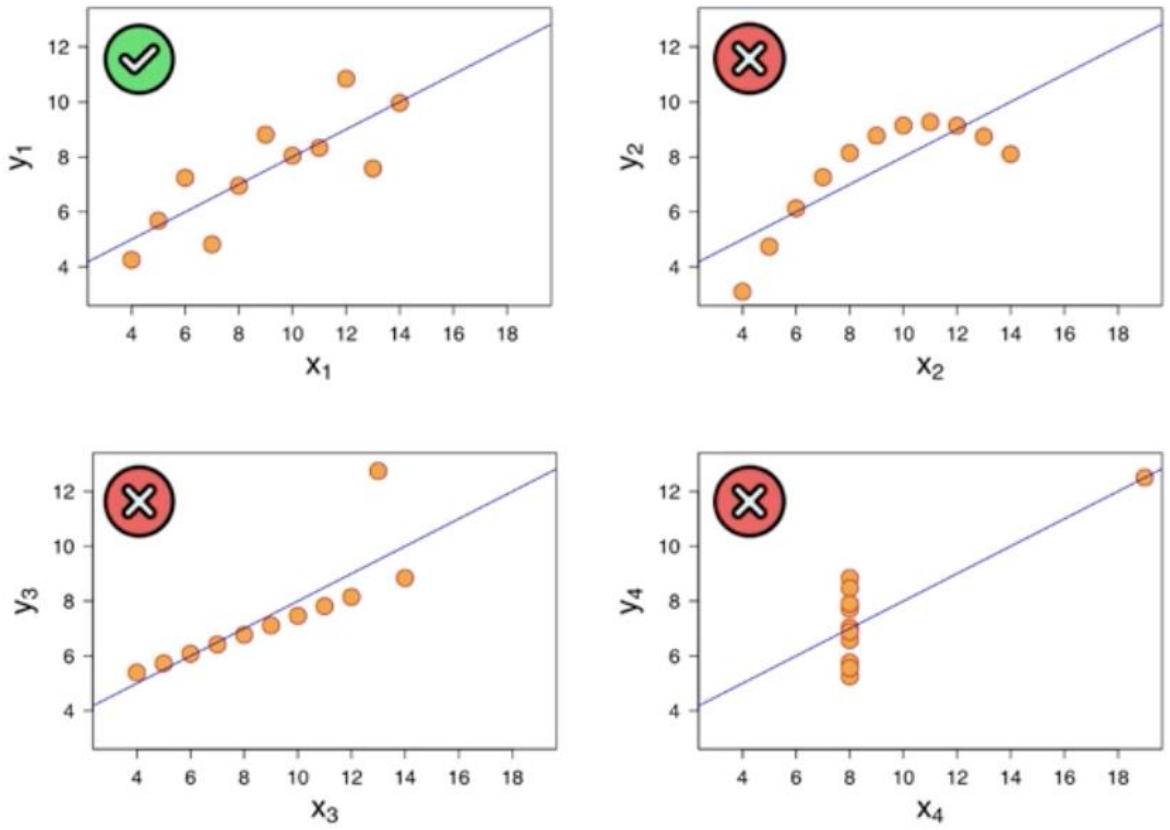
$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

Rule of thumb (for our tutorials)*:
1.0 = Perfect fit (suspicious)
~0.9 = Very good
<0.7 = Not great
<0.4 = Terrible
<0 = Model makes no sense for this data

**This is highly dependent on the context*

Linearity Assumption of Simple Linear Regression

Make sure that there is a linear relationship between the dependent variable (Y) and the independent variable (X).

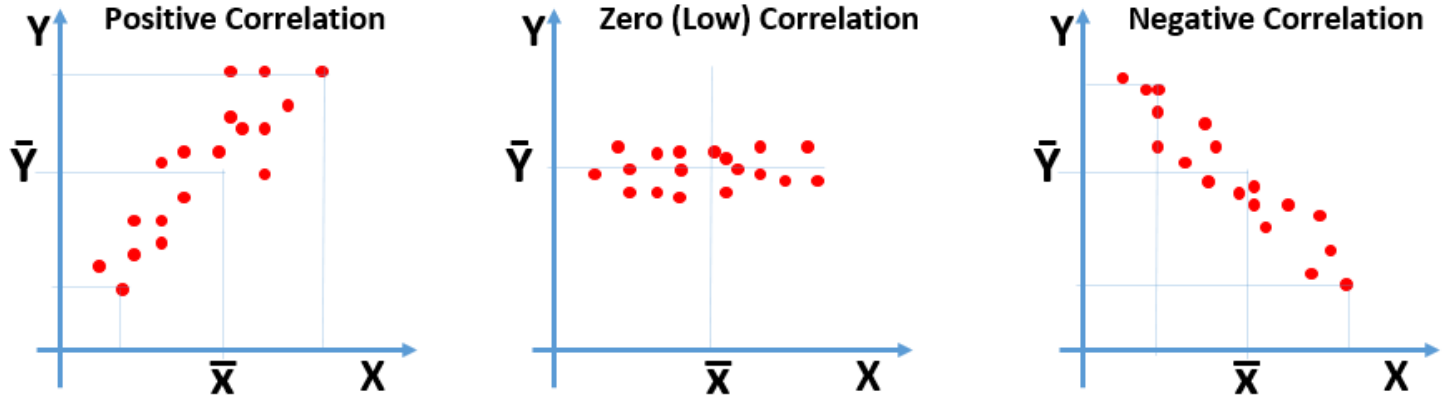


Linear correlation between two variables

Analysis of Two Continuous Variables

Scatter plot and **Correlation** are a great way of analyzing two continuous variables.

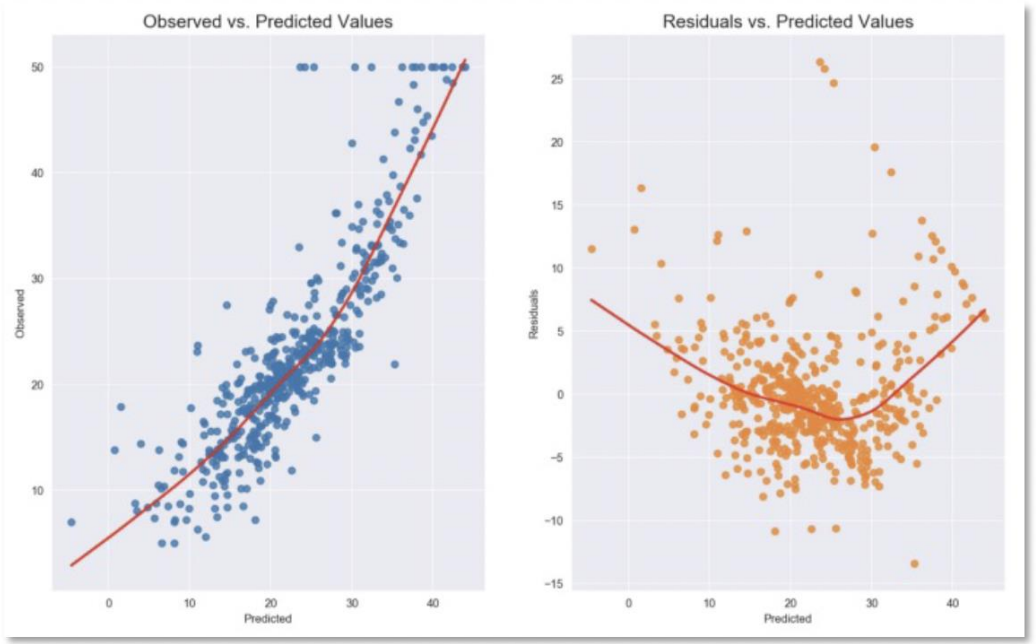
- **Correlation** measures the strength of linear relationship between Two Continuous Variables
- **Scatter Plot** visually shows the relation between the Two Continuous Variables using X – Y axis coordinate system.



https://www.k2analytics.co.in/scatter-plot-in-python/

Additional Reading

- Verifying the Assumptions of Linear Regression in Python and R, Eryk Lewinson (2019)
- towardsdatascience.com/verifying-theassumptions-of-linear-regression-in-pythonand-r-f4cd2907d4c0



Multiple Linear Regression

Multiple Linear Regression

Simple Linear Regression

$$y = b_0 + b_1 * x_1$$

Multiple Linear Regression

Dependent variable (DV) Independent variables (IVs)

$$y = b_0 + b_1 * x_1 + b_2 * x_2 + \ldots + b_n * x_n$$

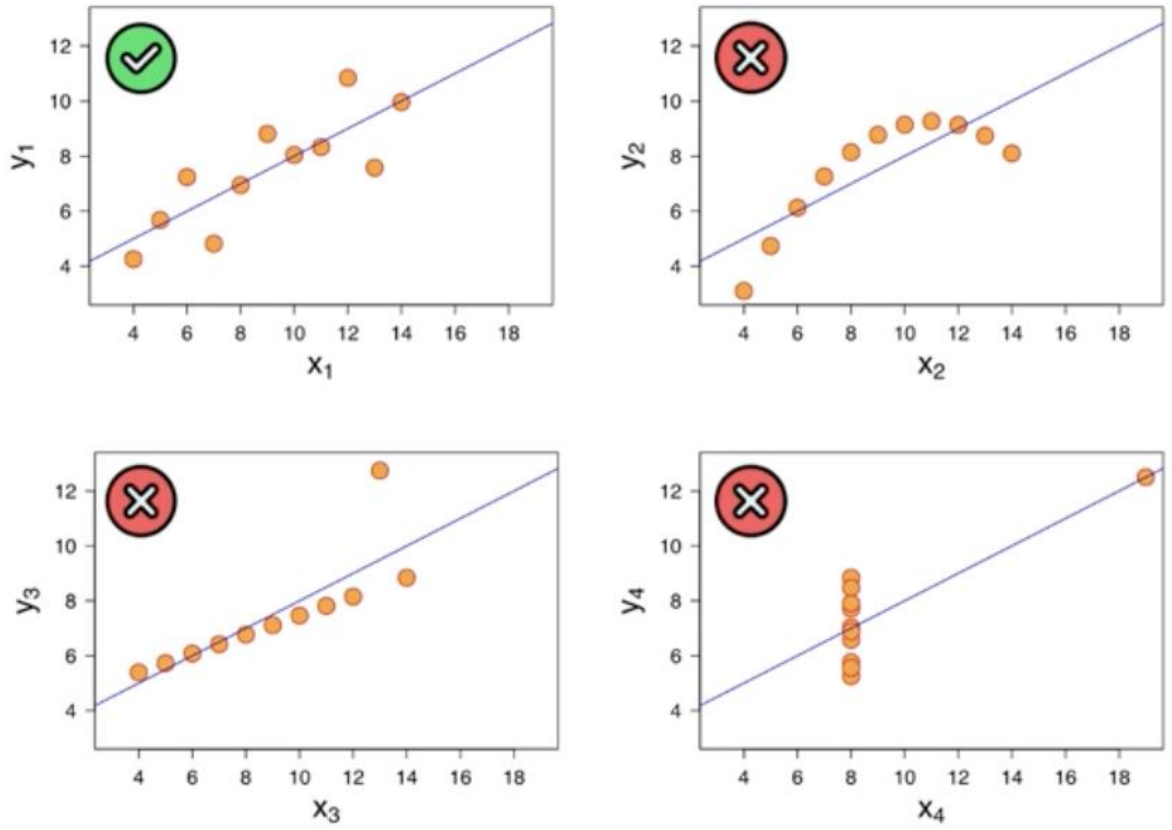
Multiple Linear Regression

Profit	R&D Spend	Admin	Marketing	State
192,261.83	165,349.20	136,897.80	471,784.10	New York
191,792.06	162,597.70	151,377.59	443,898.53	California
191,050.39	153,441.51	101,145.55	407,934.54	California
182,901.99	144,372.41	118,671.85	383,199.62	New York
166,187.94	142,107.34	91,391.77	366,168.42	California

R&D Spend	Administration	Marketing Spend	State	Profit
165349.2	136897.8	471784.1	New York	192261.83
162597.7	151377.59	443898.53	California	191792.06
153441.51	101145.55	407934.54	Florida	191050.39
144372.41	118671.85	383199.62	New York	182901.99
142107.34	91391.77	366168.42	Florida	166187.94
131876.9	99814.71	362861.36	New York	156991.12
134615.46	147198.87	127716.82	California	156122.51
130298.13	145530.06	323876.68	Florida	155752.6
120542.52	148718.95	311613.29	New York	152211.77
123334.88	108679.17	304981.62	California	149759.96
101913.08	110594.11	229160.95	Florida	146121.95
100671.96	91790.61	249744.55	California	144259.4
93863.75	127320.38	249839.44	Florida	141585.52
91992.39	135495.07	252664.93	California	134307.35
119943.24	156547.42	256512.92	Florida	132602.65
114523.61	122616.84	261776.23	New York	129917.04
78013.11	121597.55	264346.06	California	126992.93
94657.16	145077.58	282574.31	New York	125370.37
91749.16	114175.79	294919.57	Florida	124266.9

Linearity Assumption of Multiple Linear Regression

Linearity: make sure that there is a linear relationship between the dependent variable (Y) and each independent variable (Xi).



Assumptions of Multiple Linear Regression: **No Multi-collinearity**

No Multi-collinearity:

Predictors are not correlated with each other

Multi-collinearity

- Adding more independent variables to a multiple regression create more relationships among them.

- Not only the independent variables potentially related to the dependent variable, they also potentially related to each other

- The ideal is for all of the independent variables to be correlated with the dependent variable **but not with each other !**

Assumptions of Multiple Linear Regression: **No Multi-collinearity**

No Multi-collinearity:

Predictors should not be highly correlated with one another

- Avoid **Multi-collinearity** => This means that the independent variables in your model should not be highly correlated with each other.
- Solving **Multi-collinearity**:
  - **Remove a variable** — Drop the redundant one. Keep the variable that adds the most unique information.
  - **Combine variables** — Merge correlated variables into one (e.g., create a "total living space" score from size + bedrooms).
  - **Use Regularization** — Apply **Ridge** or **Lasso** regression to reduce the effect of multi-collinearity while keeping all variables in the model.

Assumptions of Multiple Linear Regression: **No Multi-collinearity**

No Multi-collinearity:

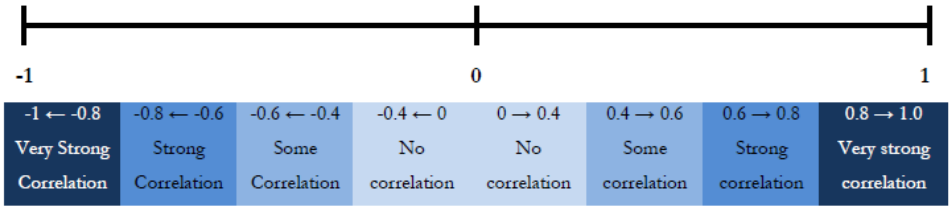
Predictors should not be highly correlated with one another

Example: Predicting House Price. Suppose you have these 3 predictors:

- X1 = House size (sq.m.)
- X2 = Number of bedrooms
- X3 = Number of bathrooms
- The problem: bigger houses tend to have more bedrooms **and** more bathrooms — so X1, X2, and X3 are all highly correlated with each other.
- **What to do:** Ask yourself — *which variable best explains house price on its own?*
- ✅ Keep **X1 (House size)** — it directly measures space and strongly predicts price.
- ❌ Drop **X2 and X3** — they don't add much new information beyond what house size already tells us.

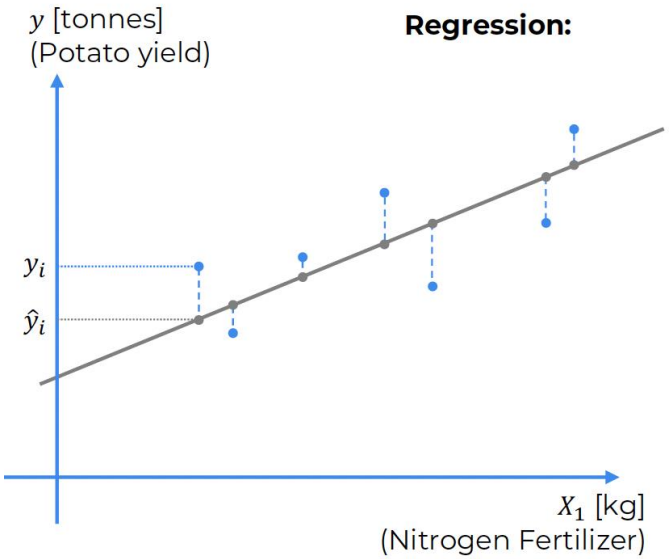
Pre-process steps for Multiple Linear Regression

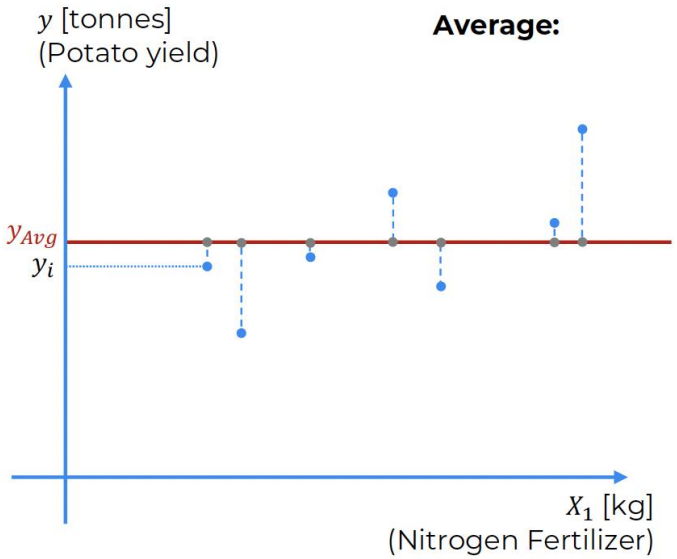
- For each independent variable, **check for linearity with the dependent variable.**
  - Scatterplot
  - Correlation



- **Check relationships among the independent variables** to eliminate co-linear features (Multi-collinearity).
  - Scatterplot
  - Correlation
- Feature subset selection

Coefficient of Determination: Adjusted r^2





$$SS_{res} = SUM(y_i - \hat{y}_i)^2$$

$$SS_{tot} = SUM(y_i - y_{avg})^2$$

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

Rule of thumb (for our tutorials)*:
1.0 = Perfect fit (suspicious)
~0.9 = Very good
<0.7 = Not great
<0.4 = Terrible
<0 = Model makes no sense for this data

**This is highly dependent on the context*

$$Adj\ R^2 = 1 - (1 - R^2) \times \frac{n - 1}{n - k - 1}$$

k – number of independent variables
n – sample size

Example: Prices of Toyota Corolla

ToyotaCorolla.xls

Goal: predict prices of used Toyota Corollas based on their specification

Data: Prices of 1442 used Toyota Corollas, with their specification information

Data Sample

(showing only the variables to be used in analysis)

Price	Age	KM	Fuel_Type	HP	Metallic	Automatic	cc	Doors	Quarterly_Tax	Weight
13500	23	46986	Diesel	90	1	0	2000	3	210	1165
13750	23	72937	Diesel	90	1	0	2000	3	210	1165
13950	24	41711	Diesel	90	1	0	2000	3	210	1165
14950	26	48000	Diesel	90	0	0	2000	3	210	1165
13750	30	38500	Diesel	90	0	0	2000	3	210	1170
12950	32	61000	Diesel	90	0	0	2000	3	210	1170
16900	27	94612	Diesel	90	1	0	2000	3	210	1245
18600	30	75889	Diesel	90	1	0	2000	3	210	1245
21500	27	19700	Petrol	192	0	0	1800	3	100	1185
12950	23	71138	Diesel	69	0	0	1900	3	185	1105
20950	25	31461	Petrol	192	0	0	1800	3	100	1185

Variables Used

Price	in Euros
Age	in months as of August 2004
KM	accumulated kilometers
Fuel Type	(diesel, petrol, CNG)
HP	horsepower
Metallic color	(1=yes, 0=no)
Automatic transmission	(1=yes, 0=no)
CC	(cylinder volume)
Doors	number of doors
Quarterly_Tax	(road tax)
Weight	(in kg)

Variables Used

Price in Euros

Age in months as of 8/04

KM (kilometers)

Fuel Type (diesel, petrol, CNG)

HP (horsepower)

Metallic color (1=yes, 0=no)

Automatic transmission (1=yes, 0=no)

CC (cylinder volume)

Doors

Quarterly_Tax (road tax)

Weight (in kg)

Steps of Predictive Modeling

Goal: predict target values

- **Model Goal:** Optimize predictive accuracy
- **Train model** on training data
- **Assess performance** on **test (hold-out) data**
- **Explaining role of predictors** is very useful

Predictive Modeling

Subset of the records selected for training partition
Categorical, must be transformed into binary variables

Price	Age	KM	Fuel_Type
13500	23	46986	Diesel
13750	23	72937	Diesel
13950	24	41711	Diesel
14950	26	48000	Diesel
13750	30	38500	Diesel
12950	32	61000	Diesel
16900	27	94612	Diesel
18600	30	75889	Diesel
21500	27	19700	Petrol
12950	23	71138	Diesel
20950	25	31461	Petrol

Id	Model	Price	Age_08_04	Mfg_Month	Mfg_Year	KM	Fuel_Type_Diesel	Fuel_Type_Petrol
4D HATCHB TE	RRA 2/3-Doors	13500	23	10	2002	46986	1	0
4D HATCHB TE	RRA 2/3-Doors	14950	26	7	2002	48000	1	0
D4D HATCHB	SOL 2/3-Doors	13750	30	3	2002	38500	1	0
D4D HATCHB	SOL 2/3-Doors	12950	32	1	2002	61000	1	0
800 T SPORT	VVT I 2/3-Doors	21500	27	6	2002	19700	0	1
D HATCHB TE	RRA 2/3-Doors	12950	23	10	2002	71138	1	0
3DR T SPORT	BNS 2/3-Doors	19950	22	11	2002	43610	0	1
VTLI 3DR T SP	ORT 2/3-Doors	22750	30	3	2002	34000	0	1

For example, 60% training data / 40% test data

The Fitted Regression Model

Input variables	Coefficient	Std. Error	p-value	SS
Constant term	-3608.418457	1458.620728	0.0137	97276410000
Age_08_04	-123.8319168	3.367589	0	8033339000
KM	-0.017482	0.00175105	0	251574500
Fuel_Type_Diesel	210.9862518	474.9978333	0.6571036	6212673
Fuel_Type_Petrol	2522.066895	463.6594238	0.00000008	4594.9375
HP	20.71352959	4.67398977	0.00001152	330138600
Met_Color	-50.48505402	97.85591125	0.60614568	596053.75
Automatic	178.1519013	212.0528565	0.40124047	19223190
cc	0.01385481	0.09319961	0.88188446	1272449
Doors	20.02487946	51.0899086	0.69526076	39265060
Quarterly_Tax	16.7742424	2.09381151	0	160667200
Weight	15.41666317	1.40446579	0	214696000

Regression evaluation: Error reports

Training Data scoring - Summary Report

Total sum of squared errors	RMS Error	Average Error
1514553377	1325.527246	-0.000426154

Validation Data scoring - Summary Report

Total sum of squared errors	RMS Error	Average Error
1021587500	1334.079894	116.3728779

Regression evaluation measures: MAE, MAPE and RMSE

- MAE (**Mean Absolute Error**) is a metric used to evaluate the average prediction error of a model, regardless of the direction of that error ความแตกต่างเฉลี่ยโดยไม่พิจารณาทิศทาง (บวกหรือลบ)
- MAE = $1/n * \Sigma |y_i - \hat{y}_i|$
  - n = number of data points
  - y_i = actual value
  - $\hat{y}_i$ = predicted value

- MAPE (**Mean Absolute Percentage Error**) is a metric used to evaluate the prediction error of a model, but unlike MAE, it expresses the error as a percentage of the actual value (percentage of errors does the model have). This can be useful when dealing with data where values have different scales.
  - MAPE = $1/n * \Sigma | (y_i - \hat{y}_i) / y_i | * 100\%$

- RMSE (**Root Mean Squared Error**) measures the average magnitude of the error between predicted and actual values. วัดขนาดของข้อผิดพลาด โดยเน้นข้อผิดพลาดใหญ่กว่าข้อผิดพลาดเล็กเนื่องจากการยกกำลังสอง
- RMSE = $\text{sqrt} (1/n * \Sigma (y_i - \hat{y}_i)^2)$

Predicted Values

Predicted price computed using regression coefficients

Predicted Value	Actual Value	Residual
15863.86944	13750	-2113.869439
16285.93045	13950	-2335.930454
16222.95248	16900	677.047525
16178.77221	18600	2421.227789
19276.03039	20950	1673.969611
19263.30349	19600	336.6965066
18630.46904	21500	2869.530964
18312.04498	22500	4187.955022
19126.94064	22000	2873.059357
16808.77828	16950	141.2217206
15885.80362	16950	1064.196384
15873.97887	16250	376.0211263
15601.22471	15750	148.7752903
15476.63164	15950	473.3683568
15544.83584	14950	-594.835836
15562.25552	14750	-812.2555172
15222.12869	16750	1527.871313
17782.33234	19000	1217.667664

Residuals = difference between actual and predicted prices

Over-fitting

- **Overfitting** occurs when your model memorizes the training data too well.
- This makes the model perform extremely well on the training data, but poorly on unseen data.
- Adding more independent variables to a multiple regression not mean the regression will be better offer better predictions
  - **Too many independent variables**: Too many variables increases the model complexity.
  - **Non-linear relationships**: If the relationships between variables are non-linear, a linear regression model might overfit by trying to create a complex linear fit, capturing noise instead of the true pattern.

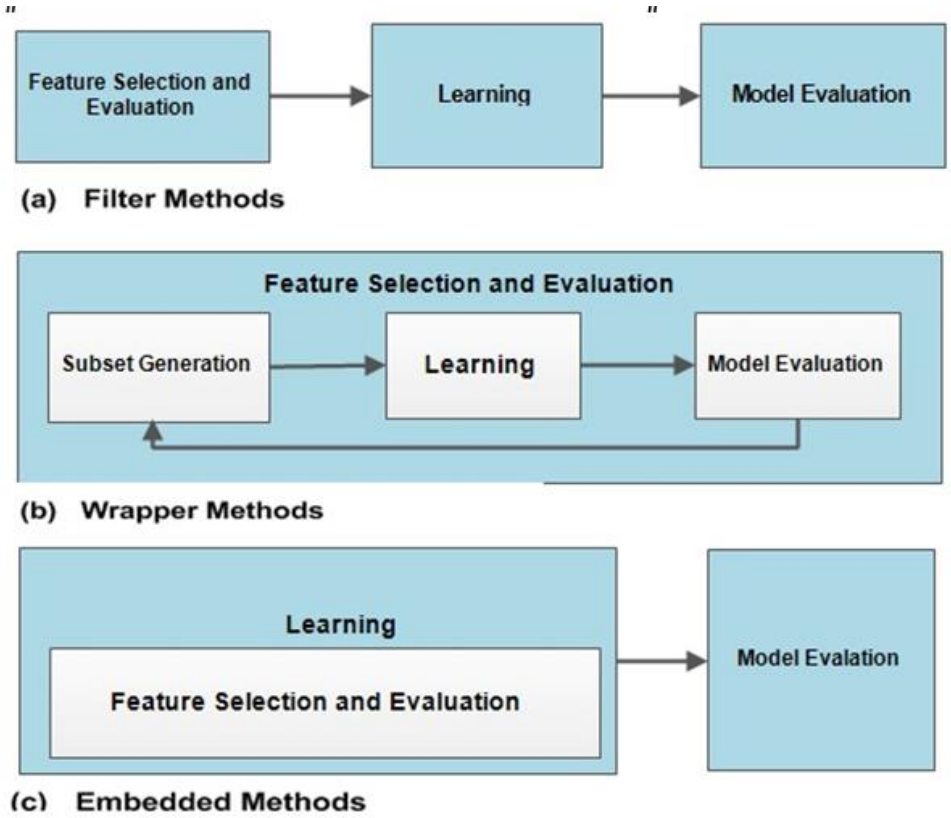
=> **Reduce number of predictors**

Reducing Number of Predictors

Goal: Find the simplest model that performs sufficiently well

- Not feasible to collect full predictors for future predictions
- Few accurate predictors
- More predictors => higher chance to missing values
- Good model is a Parsimony Model => More insight with few predictors

Feature Subset Selection



Feature Engineering

Selecting Subsets of Predictors

Goal: Find the simplest model that performs sufficiently well

- More robust
- Higher predictive accuracy

Exhaustive Search

Partial Search Algorithms

Backward/Forward Selection

- **Backward Selection**
  - Start with all predictors
  - Delete them one by one
  - Stop when the supression is not statistically significant

- **Forward Selection**
  - Start with no predictors
  - Add them one by one (add the one with largest contribution)
  - Stop when the addition is not statistically significant

Diagnostics for the 12 models

	#Coeffs	RSS	Cp	R-Squared	Adj. R-Squared
Choose Subset	2	2538203648	566.4946289	0.759902259	0.759623076
Choose Subset	3	2245803264	404.393219	0.787561455	0.787066837
Choose Subset	4	1796573056	154.2755432	0.830055744	0.829461533
Choose Subset	5	1689283456	96.06230164	0.84020465	0.839458814
Choose Subset	6	1555462272	22.9589653	0.852863273	0.85200383
Choose Subset	7	1516825984	3.27544785	0.856518017	0.855511126
Choose Subset	8	1515638144	4.60880661	0.856630379	0.855455219
Choose Subset	9	1515206272	6.36643076	0.856671232	0.855326999
Choose Subset	10	1514873088	8.1794405	0.856702749	0.855189045
Choose Subset	11	1514592768	10.02211857	0.856729265	0.855045708
Choose Subset	12	1514553344	11.99999332	0.856732995	0.854878951

Good model has: High adj-R^2

Model with only 6 predictors

The Regression Model

Input variables	Coefficient	Std. Error	p-value	SS
Constant term	-3874.492188	1415.003052	0.00640071	97276411904
Age_08_04	-123.4366303	3.33806777	0	8033339392
KM	-0.01749926	0.00173714	0	251574528
Fuel_Type_Petrol	2409.154297	319.5795288	0	5049567
HP	19.70204735	4.22180223	0.00000394	291336576
Quarterly_Tax	16.88731384	2.08484554	0	192390864
Weight	15.91809368	1.26474357	0	281026176

Model Fit →

Training Data scoring - Summary Report

Total sum of squared errors	RMS Error	Average Error
1516825972	1326.521353	-0.000143957

Predictive performance →

Validation Data scoring - Summary Report

Total sum of squared errors	RMS Error	Average Error
1021510219	1334.029433	118.4483556

Multiple Linear Regression

MULTIPLE REGRESSION MODEL

Multiple Regression Model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots \beta_p x_p + \epsilon$$

linear parameters — error

Multiple Regression Equation

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots \beta_p x_p$$

error term assumed to be zero

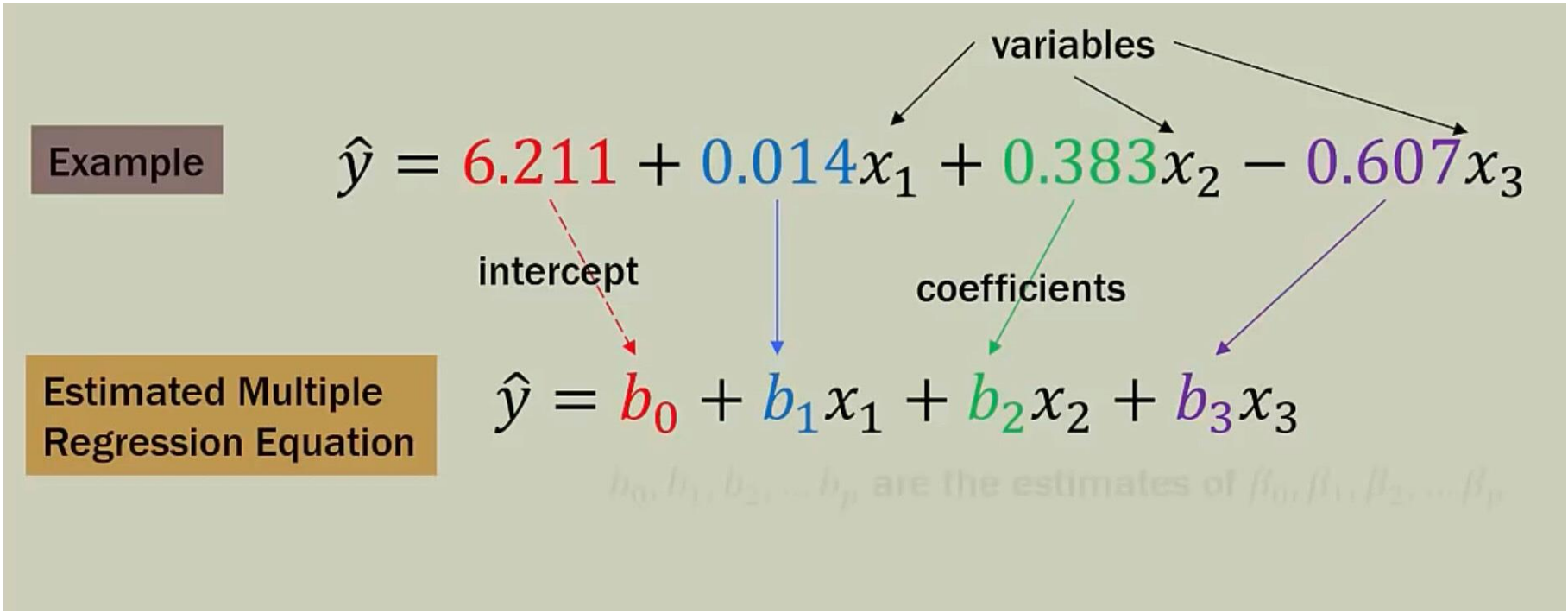
Estimated Multiple Regression Equation

$$\hat{y} = b_0 + b_1 x_1 + b_2 x_2 + \cdots b_p x_p$$

$b_0, b_1, b_2, \ldots b_p$ are the estimates of $\beta_0, \beta_1, \beta_2, \ldots \beta_p$

$\hat{y}$ = predicted value of the dependent variable

Estimated Multiple Regression Equation



Interpreting the Coefficients

$$\hat{y} = 27 + 9x_1 + 12x_2$$

x_1 = capital investment ($1000s)
x_2 = marketing expenditures ($1000s)
$\hat{y}$ = predicted sales ($1000s)

In multiple regression, each coefficient is interpreted as the estimated change in y corresponding to a one unit change in a variable, when all other variables are held constant.

So in this example, $9000 is an estimate of the expected increase in sales y, corresponding to a $1000 increase in capital investment (x_1) when marketing expenditures (x_2) are held constant.

Summary

- Linear regression models are very popular tools, not only for explanatory modeling, but also for predictive modeling.
- Predictive models are built using a training data set, and evaluated on a separate validation data set

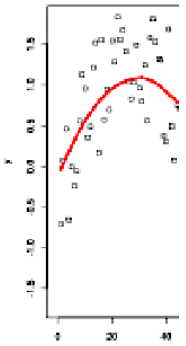
- Multiple linear regression require
- => Assumptions : **Linearity,** No-Multicollinearity, ...

- **Removing redundant predictors** is key to achieving predictive accuracy and robustness
- **Subset selection** methods help find “good” candidate models. These should then be run and assessed.

Polynomial Regression

Given: Set of known (x,y) points

Find: function f that "best fits" the known points, i.e., $f(x)$ is close to y

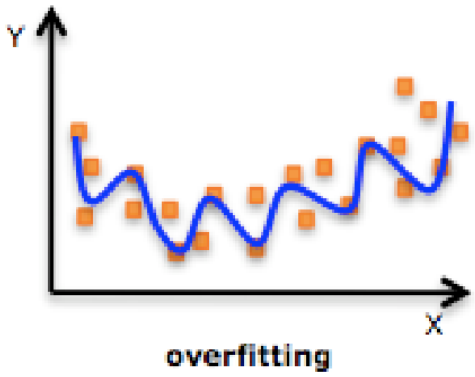


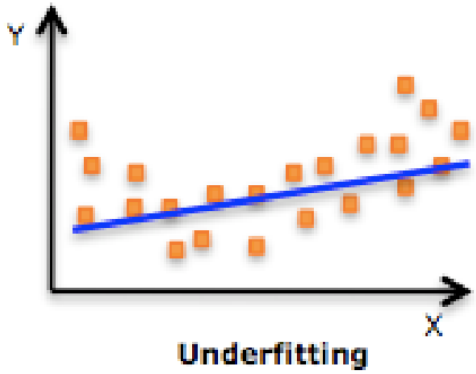
$$f(x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n$$

- "Best fit" is still method of least squares
- Still have coefficient of determination R^2
- Pick smallest degree ***n*** that fits the points reasonably well

Polynomial Regression

Overfitting and Underfitting (From Overview)







Summary

Linear regression models are very popular tools, not only for explanatory modeling, but also for predictive modeling.

Prediction models are built using a **training data set**, and evaluated on a separate **testing data set**.

Multiple linear regression require

=> Assumptions : **Linearity, No-Multicollinearity, …**

Removing redundant predictors is key to achieving predictive accuracy and robustness

Subset selection methods help find "good" candidate models. These should then be run and assessed.

Resources for further learning

- Books:
  - "Linear Regression Models" by Kutner, Nachtsheim, Neter, and Li
  - "An Introduction to Statistical Learning" by James, Witten, Hastie, and Tibshirani
- Papers:
  - "A Gentle Introduction to Linear Regression" by Gareth James
  - "Linear Regression: Assumptions and Conditions" by Theobald
  - The Application of Multiple Linear Regression and Artificial Neural Network Models for Yield Prediction *of Very Early Potato Cultivars before Harvest*, *Magdalena Piekutowska et. al. (2021)*
- Websites:
  - https://www.statmethods.net/stats/regression.html
  - https://www.mdpi.com/2073-4395/11/5/885